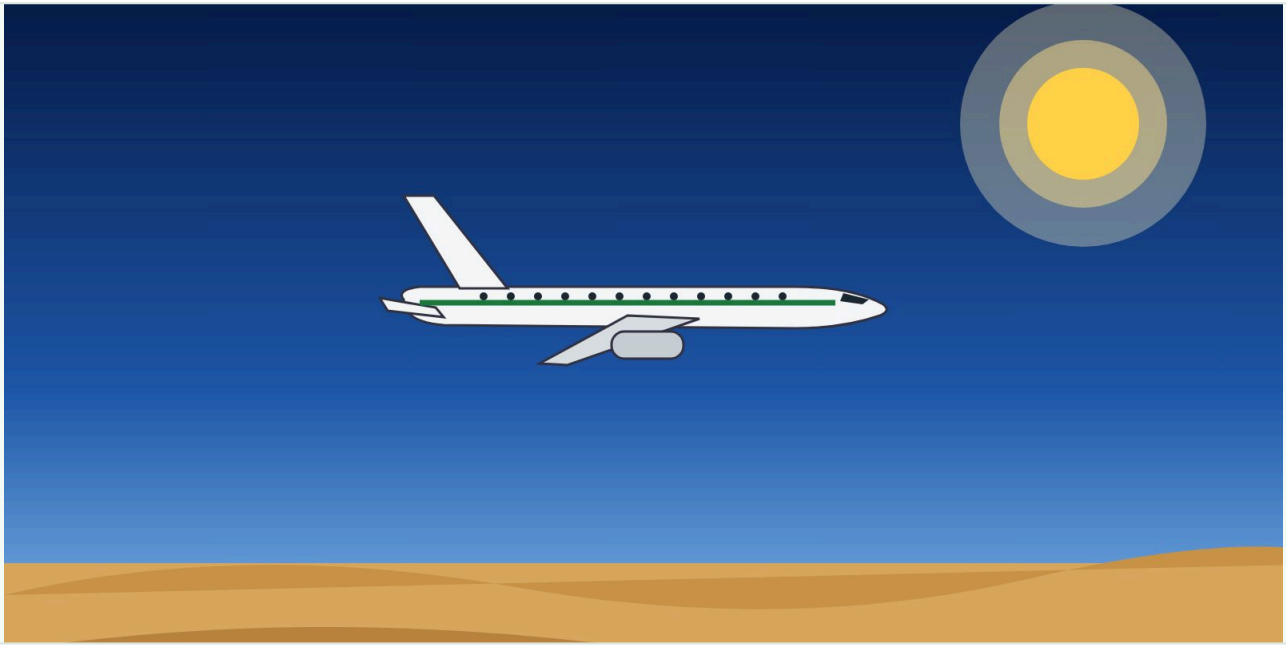
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

Lesson 03: Fuel in Hot Weather

Lesson 3 of 6 · about 6 min

LEARNING OBJECTIVES

- Explain the effect of temperature on fuel density and quantity.
- Apply fuel checks and MEL fuel items.
- Use the operational flight plan fuel figures correctly.



Hot fuel is less dense — check the uplift in kilograms.

Fuel density

- Hot fuel is **less dense**: the same mass takes up more volume. Uplift is often measured in litres but planned in kilograms — always convert with the **actual density** on the fuel docket.
- Check the **uplift**: the difference between the calculated and the actual uplift must be within the company tolerance.
- Respect any **maximum fuel temperature** limit in the AFM.

Fuel planning

- Use the Jetplan figures: trip fuel, contingency, alternate, final reserve, taxi and any extra fuel. Check the effect of a change of cruise level or take-off mass.
- Hot weather may impose a lower cruise level or a longer climb. Plan for possible weather deviations around convective activity.
- Check the zero fuel weight and fuel entries carefully — incorrect ZFW or fuel entries are a recurring flight safety event (Lesson 5).

Fuel system and MEL

- B737-800: perform the **spar fuel shut-off valve check** when required (FCOM SP 12-01).
- Read MEL fuel items in full, including the **(M) maintenance** and **(O) operational** procedures.

EXAMPLE FROM THE COMPANY LOFT — B737-800, MEL 28-01-03-01

A main tank aft fuel boost pump may be inoperative provided that: (a) both main tank forward pumps operate normally; (b) at the start of take-off, the fuel quantity in the associated tank is not less than 7,500 lb (3,402 kg); (c) a minimum fuel quantity of 2,500 lb (1,134 kg) is maintained in the associated tank; and (d) the boost pump is deactivated.

Always use the current MEL — this example is for training only.

KEY TAKEAWAYS

- Hot fuel is less dense — convert volume to mass with the actual density.
- Check the uplift and the ZFW / fuel entries carefully.
- Read every MEL item in full, including (M) and (O) procedures.

KNOWLEDGE CHECK

Why must you use the actual fuel density when checking the uplift on a hot day?

A. Fuel is heavier when hot

B. Hot fuel is less dense, so a given volume weighs less

C. Density does not matter

D. To calculate the cabin temperature

Answer: B. Hot fuel occupies more volume for the same mass; the actual density converts litres to kilograms correctly.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.